

The Definition of the Derivative



Using the limit definition

- 1 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = 3x^2 - x$.
 - 2 Use the limit definition to find $f'(x)$ for $f(x) = \frac{1}{x+2}$.
-

The derivative as a limit at a point

- 3 Use $f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$ to find $f'(3)$ for $f(x) = x^2 - 2x$.
 - 4 Use the alternate form $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ to find $f'(2)$ for $f(x) = x^3$.
-

Differentiability and graphical interpretation

- 5 Explain why $f(x) = |x - 2|$ is not differentiable at $x = 2$, computing the one-sided derivatives.
 - 6 State three conditions under which a function fails to be differentiable at a point, with a brief description of each (e.g. corner, cusp, vertical tangent, discontinuity).
-

Power, sum, and constant rules

- 7 Differentiate each function using the power rule:
a $f(x) = 7x^5$ **b** $f(x) = \frac{4}{x^2}$ **c** $f(x) = 6\sqrt[3]{x}$ **d** $f(x) = 9$
 - 8 Differentiate $f(x) = 5x^4 - 3x^3 + 2x - 7$.
-

Tangent lines from the derivative

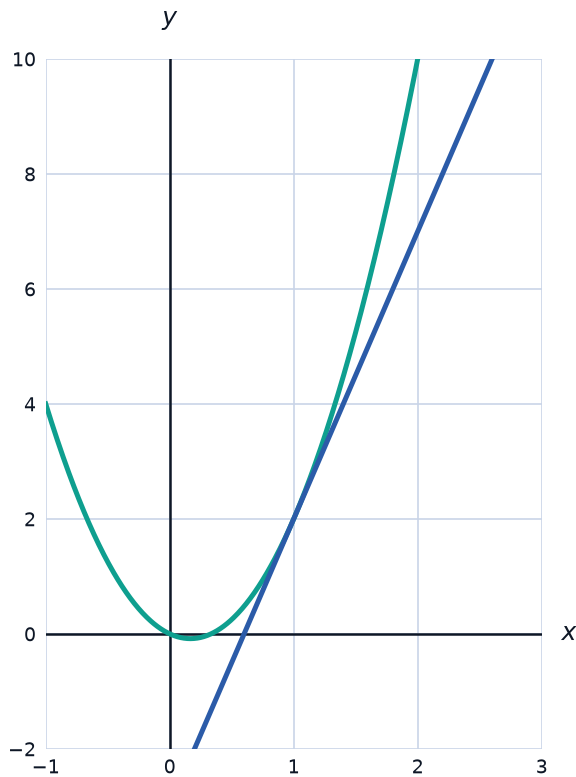
- 9 Using the limit definition result $f'(x) = 6x - 1$ found earlier for $f(x) = 3x^2 - x$, find the equation of the tangent line to f at $x = 1$.
 - 10 Find the equation of the tangent line to $f(x) = x^3$ at $x = 2$, using $f'(2) = 12$ (found earlier).
-

Estimating derivatives numerically

- 11 Given $f(1.9) = 3.42$ and $f(2.1) = 3.98$, use a symmetric difference quotient to estimate $f'(2)$.
-

Visualizing the derivative as a tangent slope

- 12 The graph shows $f(x) = 3x^2 - x$ (from an earlier problem) with its tangent line at $x = 1$, where $f'(x) = 6x - 1$.



- a Confirm the tangent line's slope matches $f'(1)$.
- b Explain what it means graphically for the line to be "tangent" at this point.
-

More practice with the limit definition

- 13 Use the limit definition to find $f'(x)$ for $f(x) = \sqrt{x + 1}$.
-

The derivative in applied motion contexts

- 14 A ball's height is $h(t) = -5t^2 + 30t + 1$ (meters, seconds). Use the limit definition of the derivative (or the power rule, since it agrees) to find $h'(t)$, then find the ball's velocity at $t = 2$.

- 15 For the same height function, find the time at which the ball's velocity is 0 (its highest point).
-

Comparing average and instantaneous rate of change

- 16 For $f(x) = x^2$, find the average rate of change on $[1, 3]$, then compare it to the instantaneous rate of change $f'(2)$ at the midpoint $x = 2$.
-

Multiple choice — easy 15

- 17 For $f(x) = 3x^2 - 4x - 1$, use the power rule to find $f'(2)$.
- a. 2
 - b. 6
 - c. 8
 - d. 10
- 18 For $f(x) = 5x^2 - 5x + 4$, use the power rule to find $f'(3)$.
- a. 10
 - b. 25
 - c. 27
 - d. 23
- 19 For $f(x) = 3x^2 + 4x + 6$, use the power rule to find $f'(-1)$.
- a. -4
 - b. 0
 - c. 1
 - d. -2

20 For $f(x) = 4x^2 - 8x - 4$, use the power rule to find $f'(-2)$.

a. -22

b. -16

c. -24

d. -26

21 For $f(x) = 5x^2 - 4x - 2$, use the power rule to find $f'(2)$.

a. 6

b. 16

c. 18

d. 14

22 For $f(x) = x^2 - 9x + 3$, use the power rule to find $f'(-3)$.

a. -12

b. -13

c. -17

d. -15

23 For $f(x) = 2x^2 - 6x - 9$, use the power rule to find $f'(3)$.

a. 6

b. 8

c. 0

d. 4

24 For $f(x) = 6x^2 + 9x - 5$, use the power rule to find $f'(3)$.

a. 43

b. 27

c. 45

d. 47

25 For $f(x) = 6x^2 + 4x + 4$, use the power rule to find $f'(0)$.

a. 4

b. 6

c. 2

d. 5

26 For $f(x) = 3x^2 + 3x + 4$, use the power rule to find $f'(-4)$.

a. -21

b. -19

c. -23

d. -9

27 For $f(x) = 5x^2 + 4x + 4$, use the power rule to find $f'(0)$.

a. 6

b. 5

c. 2

d. 4

28 For $f(x) = 3x^2 - 9x + 3$, use the power rule to find $f'(-4)$.

- a. -31
- b. -35
- c. -33
- d. -21

29 For $f(x) = 5x^2 + 3x - 6$, use the power rule to find $f'(0)$.

- a. 5
- b. 1
- c. 3
- d. 4

30 For $f(x) = x^2$, use the power rule to find $f'(3)$.

- a. 4
- b. 3
- c. 6
- d. 8

31 For $f(x) = 2x^2 - 6x - 2$, use the power rule to find $f'(-2)$.

- a. -14
- b. -12
- c. -16
- d. -10

Multiple choice — medium 15

- 32 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = x^2 - 2x + 5$.
- a. $2x - 3$
 - b. $x - 2$
 - c. $2x - 2$
 - d. $2x + 2$
- 33 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = x^2 - 6x + 8$.
- a. $2x - 7$
 - b. $2x - 6$
 - c. $x - 6$
 - d. $2x + 6$
- 34 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = x^2 - 6x$.
- a. $x - 6$
 - b. $2x - 7$
 - c. $2x - 6$
 - d. $2x + 6$
- 35 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = 2x^2 + x + 4$.
- a. $2x + 1$
 - b. $4x - 1$
 - c. $4x + 1$
 - d. $4x + 2$

- 36 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = 2x^2 - x - 4$.
- a. $2x - 1$
 - b. $4x - 1$
 - c. $4x + 1$
 - d. $4x - 2$
- 37 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = 6x^2 + 3x + 2$.
- a. $6x + 3$
 - b. $12x - 3$
 - c. $12x + 3$
 - d. $12x + 4$
- 38 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = 4x^2 - 3x + 2$.
- a. $8x + 3$
 - b. $4x - 3$
 - c. $8x - 4$
 - d. $8x - 3$
- 39 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = 4x^2 + 9x - 7$.
- a. $4x + 9$
 - b. $8x + 9$
 - c. $8x - 9$
 - d. $8x + 10$

- 40 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = x^2 - 5x$.
- a. $2x - 6$
 - b. $2x - 5$
 - c. $2x + 5$
 - d. $x - 5$
- 41 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = 6x^2 + x + 9$.
- a. $12x + 2$
 - b. $12x + 1$
 - c. $6x + 1$
 - d. $12x - 1$
- 42 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = 5x^2 - 4x - 9$.
- a. $10x + 4$
 - b. $10x - 4$
 - c. $5x - 4$
 - d. $10x - 5$
- 43 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = 5x^2 - 9$.
- a. $15x$
 - b. $10x + 1$
 - c. $10x$
 - d. $5x$

- 44 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = x^2 - 5x - 6$.
- a. $x - 5$
 - b. $2x - 6$
 - c. $2x + 5$
 - d. $2x - 5$
- 45 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = 3x^2 - 7x + 2$.
- a. $6x + 7$
 - b. $6x - 7$
 - c. $6x - 8$
 - d. $3x - 7$
- 46 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = 6x^2 - 2x + 4$.
- a. $12x - 2$
 - b. $6x - 2$
 - c. $12x + 2$
 - d. $12x - 3$

Multiple choice — hard 15

- 47 Use the limit definition to find $f'(x)$ for $f(x) = \frac{1}{x+6}$, then evaluate $f'(0)$.
- a. $-\frac{1}{6}$
 - b. $-\frac{1}{36}$
 - c. $-\frac{1}{37}$
 - d. $\frac{1}{36}$

48 Use the limit definition to find $f'(x)$ for $f(x) = \frac{1}{x+6}$, then evaluate $f'(4)$.

a. $-\frac{1}{100}$

b. $-\frac{1}{10}$

c. $\frac{1}{100}$

d. $-\frac{1}{101}$

49 Use the limit definition to find $f'(x)$ for $f(x) = \frac{1}{x+1}$, then evaluate $f'(-3)$.

a. $\frac{1}{2}$

b. $-\frac{1}{5}$

c. $-\frac{1}{4}$

d. $\frac{1}{4}$

50 Use the limit definition to find $f'(x)$ for $f(x) = \frac{1}{x+5}$, then evaluate $f'(-2)$.

a. $-\frac{1}{10}$

b. $-\frac{1}{9}$

c. $\frac{1}{9}$

d. $-\frac{1}{3}$

51 Use the limit definition to find $f'(x)$ for $f(x) = \frac{1}{x+2}$, then evaluate $f'(2)$.

a. $-\frac{1}{4}$

b. $\frac{1}{16}$

c. $-\frac{1}{17}$

d. $-\frac{1}{16}$

52 Use the alternate form $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ to find $f'(-2)$ for $f(x) = x^3 + 6x$.

- a. 6
- b. 0
- c. 20
- d. 18

53 Use the alternate form $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ to find $f'(4)$ for $f(x) = x^3 - 3x$.

- a. 51
- b. 45
- c. 47
- d. 9

54 Use the alternate form $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ to find $f'(3)$ for $f(x) = x^3 - 2x$.

- a. 29
- b. 25
- c. 27
- d. 7

55 Use the alternate form $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ to find $f'(-2)$ for $f(x) = x^3 - 4x$.

- a. 10
- b. 8
- c. -10
- d. 16

- 56 Use the alternate form $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ to find $f'(-3)$ for $f(x) = x^3 + 3x$.
- a. 32
 - b. 24
 - c. 30
 - d. -6
- 57 For $f(x) = x^2$ when $x \leq 0$, and $f(x) = 5x$ when $x > 0$ (chosen so f is continuous at $x = 0$), determine whether f is differentiable at $x = 0$.
- a. the left-hand derivative is $2(0) = 0$ and the right-hand derivative is 5 — they agree, so f is differentiable at $x = 0$.
 - b. f is differentiable at $x = 0$ automatically, since it is continuous there.
 - c. the left-hand derivative is $2(0) = 0$ and the right-hand derivative is 5 — they disagree, but differentiability doesn't require one-sided derivatives to match.
 - d. the left-hand derivative is $2(0) = 0$ and the right-hand derivative is 5 — they disagree, so f is not differentiable at $x = 0$ (the graph has a corner).
- 58 For $f(x) = x^2$ when $x \leq 3$, and $f(x) = 5x - 6$ when $x > 3$ (chosen so f is continuous at $x = 3$), determine whether f is differentiable at $x = 3$.
- a. f is differentiable at $x = 3$ automatically, since it is continuous there.
 - b. the left-hand derivative is $2(3) = 6$ and the right-hand derivative is 5 — they disagree, so f is not differentiable at $x = 3$ (the graph has a corner).
 - c. the left-hand derivative is $2(3) = 6$ and the right-hand derivative is 5 — they agree, so f is differentiable at $x = 3$.
 - d. the left-hand derivative is $2(3) = 6$ and the right-hand derivative is 5 — they disagree, but differentiability doesn't require one-sided derivatives to match.

- 59 For $f(x) = x^2$ when $x \leq 2$, and $f(x) = x + 2$ when $x > 2$ (chosen so f is continuous at $x = 2$), determine whether f is differentiable at $x = 2$.
- a. f is differentiable at $x = 2$ automatically, since it is continuous there.
 - b. the left-hand derivative is $2(2) = 4$ and the right-hand derivative is 1 — they agree, so f is differentiable at $x = 2$.
 - c. the left-hand derivative is $2(2) = 4$ and the right-hand derivative is 1 — they disagree, so f is not differentiable at $x = 2$ (the graph has a corner).
 - d. the left-hand derivative is $2(2) = 4$ and the right-hand derivative is 1 — they disagree, but differentiability doesn't require one-sided derivatives to match.
- 60 For $f(x) = x^2$ when $x \leq -1$, and $f(x) = -2x - 1$ when $x > -1$ (chosen so f is continuous at $x = -1$), determine whether f is differentiable at $x = -1$.
- a. the left-hand derivative is $2(-1) = -2$ and the right-hand derivative is -2 — they disagree, so f is not differentiable at $x = -1$.
 - b. the left-hand derivative is $2(-1) = -2$ and the right-hand derivative is -2 — they agree, so f is differentiable at $x = -1$.
 - c. the left-hand derivative is $2(-1) = -2$ and the right-hand derivative is -2 — they agree, but f is not differentiable since the pieces use different formulas.
 - d. f is differentiable at $x = -1$ automatically, since it is continuous there.
- 61 For $f(x) = x^2$ when $x \leq 2$, and $f(x) = 4x - 4$ when $x > 2$ (chosen so f is continuous at $x = 2$), determine whether f is differentiable at $x = 2$.
- a. the left-hand derivative is $2(2) = 4$ and the right-hand derivative is 4 — they agree, so f is differentiable at $x = 2$.
 - b. the left-hand derivative is $2(2) = 4$ and the right-hand derivative is 4 — they agree, but f is not differentiable since the pieces use different formulas.
 - c. the left-hand derivative is $2(2) = 4$ and the right-hand derivative is 4 — they disagree, so f is not differentiable at $x = 2$.
 - d. f is differentiable at $x = 2$ automatically, since it is continuous there.